MILITARY SPECIFICATION

APPLICATOR SET, PIN RETENTION, DENTAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an applicator set (spiral instruments) suitable for depositing cement into amalgam reinforcement pin holes in tooth structure.

2. APPLICABLE DOCUMENTS

2.1 Specifications, standards, and drawings. The following documents, of the issue in effect on date of invitations for bid or requests for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-W-470	Wire, Steel, High Carbon, Spring, Bright, Music.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-P-116	Preservation, Methods of.

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STANDARDS

Federal

FED-STD-66	Steel: Chemical Composition and Hardenability.
FED-STD-151	Metals; Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

24113	Applicator Set, Pin Retention, Dental

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The shank shall be fabricated of carbon steel or tool steel, and shall be bright nickel plated, or chromium over nickel plated. As an alternate, the shank shall be fabricated of AISI 302, 303, or 304 stainless steel in accordance with FED-STD-66. The operative end shall be fabricated of high carbon steel piano wire in accordance with QQ-W-470.

3.2 Style, design, and dimensions. The applicators shall be spiral filling instruments of the Lentulo design, and shall consist of an operative end securely inserted into the shank. Applicators shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 24113 for style, design, and dimensions, and shall be furnished as a set of one each of the four sizes shown in the drawing: long large, long fine, short large, and short fine. The applicators shall be suitable for use in latch type contra-angle handpieces. The large size applicators shall be capable of depositing cement upon the walls of amalgam reinforcement pin holes 0.027 inch in diameter; the fine size applicators shall be used for walls of pin holes 0.024 inch in diameter during Markley pin-retention techniques, without the entrapment of air.

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3.3 Tensile strength. Piano wire shall have a minimum tensile strength of 340,000 pounds per square inch prior to forming, when tested as specified in 4.4.1.

3.4 Identification marking. The spiral applicators shall be permanently and legibly marked with the name or registered trademark of the contractor or manufacturer. In addition, applicators of foreign manufacture shall be marked with the name of the country of origin.

3.5 Finish. The operative end and shank shall have a bright finish.

3.6 Workmanship. Workmanship shall be first class throughout. Applicators shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties.)

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the name of the material, the lot number, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in tables I and II (see 4.2.1 and 4.3) and 4.2.2.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. Unit of product for examination purposes shall be the individual applicator.

Table I. Inspection and Acceptable Quality Levels

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests on raw materials. A sufficient quantity from each lot of metal used in the construction of the applicators shall be selected for chemical analysis. A minimum of three specimens from each lot of steel shall be selected for tensile strength determination.

4.2.3 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The applicators shall be examined to determine compliance with all requirements contained in the specification. Examination shall be conducted as specified in 4.2.1. Any applicator in the sample containing one or more defects shall be rejected, and if the number of defective applicators in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Applicators shall be examined for defects including, but not limited to, those listed in table II.

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Table II. Classification of Defects

Categories	Defects
Major A:	
101	Operative ends have kinks, cracks, or other surface imperfections.
102	Latch end of shank not cleanly machined.
103	Assortment of applicators in set not furnished as specified.
Major B:	
151	Spiral diameters not as specified.
152	Shank diameter not as specified.
153	Design and dimensions of latch end of shank not as specified.
Minor A:	
201	Identification marking not as specified.
202	Packaging, packing,and marking not as specified.
203	Number of applicators in unit package not as specified.
204	Finish of applicators not as specified.
Minor B:	
251	Dimensions of applicators (other than Major B) not as specified.

4.4 Tests.

4.4.1 Tensile strength. Tests for tensile strength shall be conducted on samples of each lot of steel stock in accordance with method 211 of FED-STD-151. The testing machine shall be operated at a uniform rate of speed, which shall move the head or jaws apart at the rate of one inch per minute.

4.4.2 Chemical analysis. In the absence of certificates of quality for the raw material, samples selected in accordance with 4.2.2 shall be analyzed in conformance with methods 111 or 112 of FED-STD-151. In case of dispute, method 111 shall be used for referee analysis.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or C, as specified (see 6.1).

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5.1.1 Unit of issue. One set (4 applicators), as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table III shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table III. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	144 units

5.1.2.1 Variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall consist of a plastic tray with a plastic slipover cover or a plastic box with a hinged cover and a plastic tray. The plastic tray shall be designed with four cavities, so that four applicators can be packaged without cushioning. The cavities shall be molded, so as to hold the applicators in place in such a manner that no free movement is present to damage the operative ends of the applicators. The cover, when placed, or closed over the tray, shall be flush on the applicators. Cover shall be secured with tape. Each unit shall be further packed in accordance with MIL-P-116, method IA-8.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers, of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements .

SPECIFICATION

MIL-A-36644 FSN:

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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